

Question

The element **X** exhibits stable properties with strong corrosion resistance, capable of withstanding corrosion from hydrochloric acid, sulfuric acid, nitric acid, and aqua regia at room temperature. Fusing the elemental **X** with potassium nitrate and potassium hydroxide yields **A**. After cooling the melt, it is extracted with water and treated with chlorine gas, followed by heating to volatilize **B**. Introducing **B** into a hydrochloric acid solution and heating it, then adding ammonium chloride precipitates element **X** in the form of **C**, which, upon ignition in hydrogen gas, yields spongy elemental **X**. In a hydrochloric acid solution of potassium chloride, reducing **B** with ethanol produces the red crystalline compound **D**, in which the mass fraction of **X** is 26.99%. Heating **D** to 200 °C results in a weight loss of 4.81%, yielding the brown crystalline compound **E**. If **B** is replaced with **A** as the reactant, a deep-red diamagnetic dinuclear complex **F** is obtained, in which the mass fraction of **X** is 27.73%. Dissolving **F** in concentrated hydrochloric acid and adding ammonia water generates a 1:3 ionic compound **G**. Exposing the solution of **G** to air produces a vivid red color, from which the diamagnetic compound **H** can be isolated. **H** is named for its intense color. Oxidizing **H** with ammonium cerium(IV) sulfate yields the paramagnetic brown compound **I**. It is known that both **H** and **I** are trinuclear complexes with identical cationic chemical formulas, where the mass fraction of **X** in the cation is 52.86%, while the anion count ratio is 6:7.

Given that **A~I** are all compounds containing **X**, which of the following options are correct?

- A.** In **A**, the **X** element has a valence of +4.
- B.** **B** is a regular octahedral structure.
- C.** **C** is a 2:1 type ionic compound
- D.** **E** has a molecular weight of approximately 392g/mol
- E.** There is only one set of three-center π bonds in **F**

F. The ratio of nitrogen atoms to chlorine atoms in **H** is 7 : 3.

G. In **I**, the average oxidation state of **X** is +3

H. None of the above options are correct

Answer

Correct Answer: **F**

Detailed Explanation

Based on the fact that the elemental form of **X** can withstand corrosion by hydrochloric acid, sulfuric acid, nitric acid, and aqua regia at room temperature, and that it can form complexes, it is determined to be a high-period platinum-group metal.

CHECKPOINT

1 PTS

X is a high-period platinum-group metal

The elemental form of **X**, when fused with potassium nitrate and potassium hydroxide, yields **A**, which takes the form of K_aXO_b . After treatment with chlorine gas, volatile **B** is obtained, indicating that it is a high-valent chloride or oxide of **X**. When **B** is reduced with ethanol in a hydrochloric acid solution of potassium chloride, **D** is produced. The cation of **D** is K^+ , while the anion is a chloride or aqua complex of **X**.

When **D** is heated at 200°C , it loses 4.81% of its weight, which is attributed to the loss of water of crystallization. Assuming y water molecules are lost, when $y = 1$, the relative molecular mass of **D** can be deduced. Subsequently, based on the mass fraction of **X** in **D** being 26.99%, the relative atomic mass of **X** is calculated.

When $y = 1$, the relative atomic mass of **X** is calculated to be 101.1, corresponding to *Ru*, which fits the context.

For higher values of y , no matching element is found.

Thus, it is concluded that **X** is *Ru*, **A** is K_2RuO_4 , **B** is RuO_4 , **D** is $K_2[RuCl_5(H_2O)]$, and **E** is K_2RuCl_5 .

A is K_2RuO_4 , where the oxidation state of *Ru* is +6, making option A incorrect. **B** is RuO_4 , which has a tetrahedral structure, making option B incorrect. **E** is K_2RuCl_5 , so its molecular weight is approximately

356.6 g/mol , making option D incorrect.

CHECKPOINT

1 PTS

X is Ru

CHECKPOINT

1 PTS

A is K_2RuO_4 , where Ru is in the +6 oxidation state; option A is incorrect

CHECKPOINT

1 PTS

B is RuO_4 , with a tetrahedral structure; option B is incorrect

CHECKPOINT

1 PTS

D is $K_2[RuCl_5(H_2O)]$

CHECKPOINT

1 PTS

E is K_2RuCl_5 , with a molecular weight of approximately 356.6 g/mol ; option D is incorrect

When **B** is passed into a hydrochloric acid solution and heated, followed by the addition of ammonium chloride, **X** precipitates as **C**, which is the complex $(NH_4)_3RuCl_6$, a 3:1 ionic compound, making option C incorrect.

CHECKPOINT

1 PTS

C is the complex $(NH_4)_3RuCl_6$, a 3:1 ionic compound; option C is incorrect

F is a deep-red, diamagnetic binuclear complex with a **X** mass fraction of 27.73%. Calculations show that it contains an oxo bridge, and **F** is identified as $K_4[Ru_2OCl_{10}]$. It has two sets of three-center π bonds perpendicular to the $Ru - O - Ru$ direction, making option E incorrect.

CHECKPOINT

1 PTS

F is $K_4[Ru_2OCl_{10}]$, with two sets of three-center π bonds perpendicular to the $Ru - O - Ru$ direction; option E is incorrect

When **F** is dissolved in concentrated hydrochloric acid and treated with ammonia, a 1:3 ionic compound **G** is formed, with Cl^- as the anion. The cation is a trivalent ammonia complex, so **G** is $[Ru(NH_3)_6]Cl_3$.

CHECKPOINT

1 PTS

G is $[Ru(NH_3)_6]Cl_3$

When the solution of **G** is exposed to air, a diamagnetic compound **H** can be isolated. Oxidizing **H** with ammonium cerium(IV) sulfate yields a paramagnetic brown compound **I**. Both **H** and **I** are trinuclear complexes with identical cationic formulas. The mass fraction of **X** in the cation is 52.86%, allowing the molar ratio of the cation to be

deduced. Apart from Ru , the cation may contain Cl^- , NH_3 , or O^{2-} . Ultimately, the cation is determined to be $[Ru_3O_2(NH_3)_{14}]^+$, while the ratio of Cl^- anions is 6:7 to maintain reasonable oxidation states for **Ru**. Thus, **H** is $[Ru_3O_2(NH_3)_{14}]Cl_6$, and **I** is $[Ru_3O_2(NH_3)_{14}]Cl_7$. In **H**, the ratio of nitrogen to chlorine atoms is 7 : 3, making option F correct. In **I**, the average oxidation state of Ru is $+\frac{11}{3}$, making option G incorrect.

CHECKPOINT

1 PTS

H is $[Ru_3O_2(NH_3)_{14}]Cl_6$, with a nitrogen-to-chlorine ratio of 7 : 3; option F is correct

CHECKPOINT

1 PTS

I is $[Ru_3O_2(NH_3)_{14}]Cl_7$, with an average Ru oxidation state of $+\frac{11}{3}$; option G is incorrect

In conclusion, the correct answer for this question is F.